



US012410647B2

(12) **United States Patent**
Weaver

(10) **Patent No.:** **US 12,410,647 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **DOOR STOP MECHANISM** 6,042,158 A * 3/2000 Horn E05D 13/006
292/DIG. 36
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(*) Notice: Subject to any disclaimer, the term of this 2003/0051409 A1 3/2003 Bruns
patent is extended or adjusted under 35 2019/0153760 A1 * 5/2019 Royse E05D 13/1238
U.S.C. 154(b) by 358 days.

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(21) Appl. No.: **17/963,399** DE 202016005120 U1 * 10/2016
EP 0149692 A1 * 1/1984
(22) Filed: **Oct. 11, 2022** EP 3748112 A1 * 12/2020 E05D 13/006
EP 3608493 B1 2/2022
(65) **Prior Publication Data** WO 2020260125 A1 12/2020

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(51) **Int. Cl.** International Search Report and Written Opinion mailed in corre-
E05D 13/00 (2006.01) sponding PCT Application No. PCT/US2022/046235 on Jan. 18,
(52) **U.S. Cl.** 2023.

CPC **E05D 13/006** (2013.01); **E05D 13/1215**
(2013.01); **E05D 13/1223** (2013.01); **E05Y**
2201/47 (2013.01)

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(58) **Field of Classification Search**
CPC E05D 13/1215; E05D 13/1223; E05D
13/003; E05D 13/006; E05Y 2201/47
See application file for complete search history.

(57) **ABSTRACT**

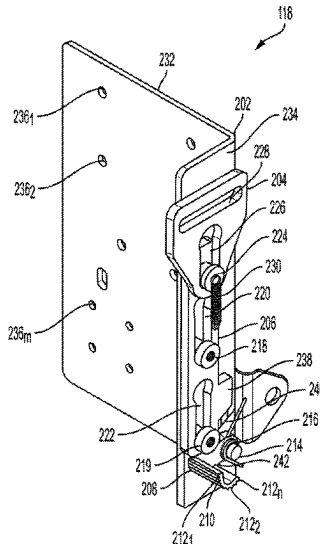
An example door stop mechanism is provided. The door stop mechanism includes a body, a connection member, a spring, and a brake. The spring is coupled to the body and the connection member. The brake is positioned adjacent to the body. The brake moves between an engaged position via a spring force of the spring that causes the body to interact with the brake such that the brake contacts a track of an overhead door and an disengaged position via a force that acts against the spring force to pull the body away from the brake such that the brake moves away from the track to allow the overhead door to move.

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18 Claims, 9 Drawing Sheets



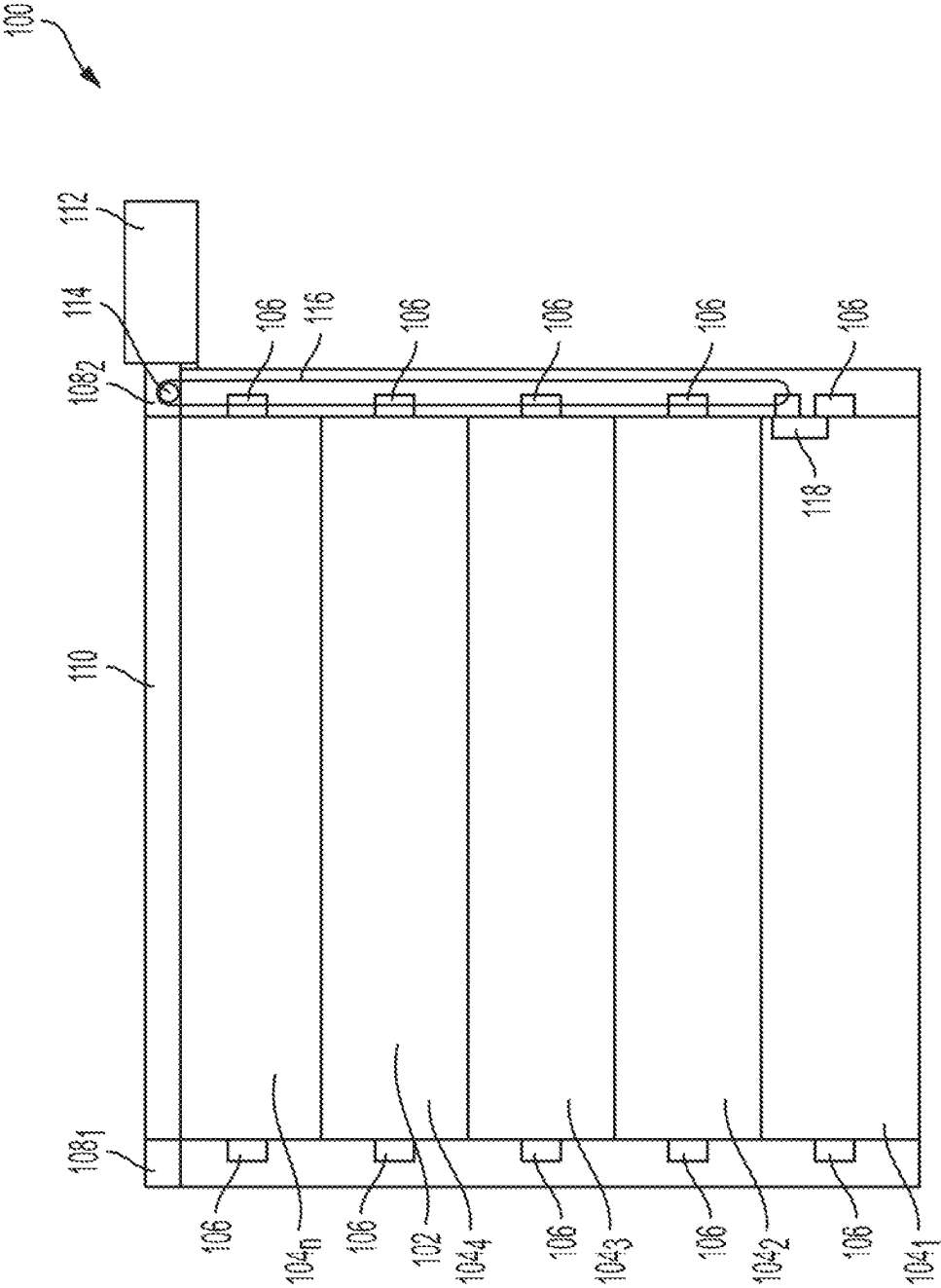


FIG. 1

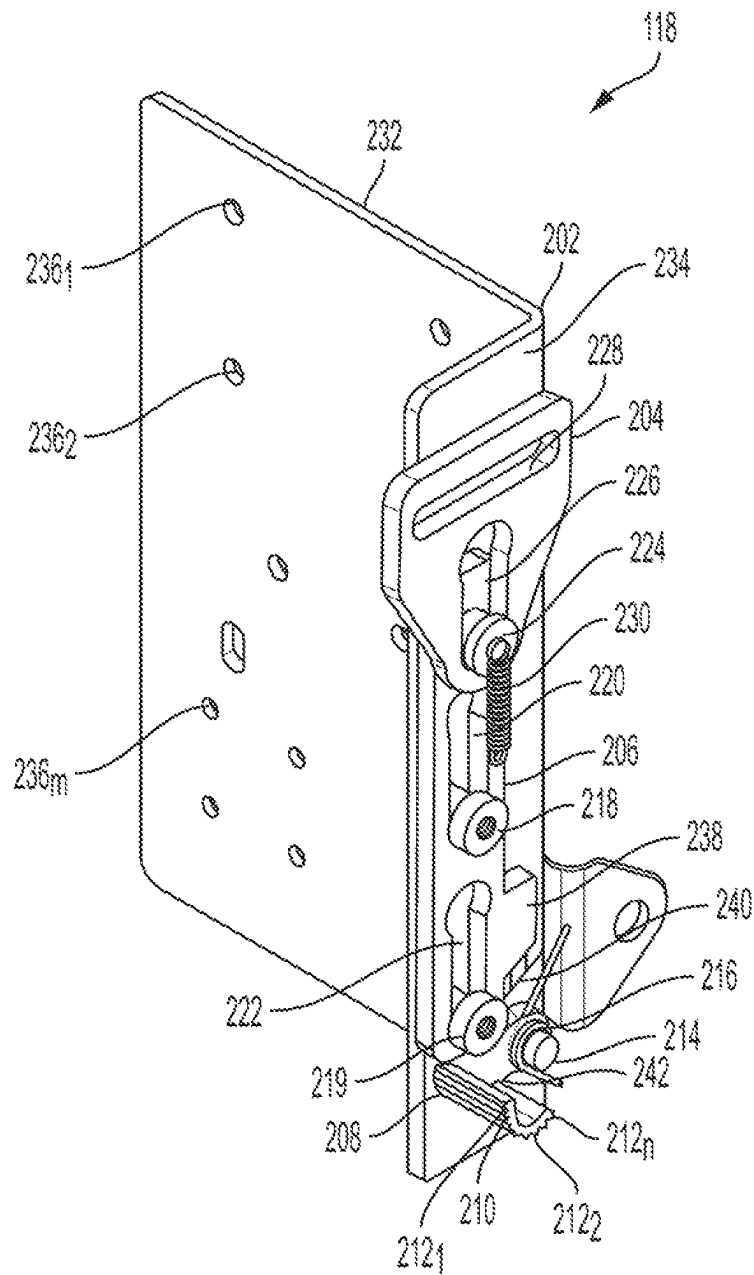


FIG. 2

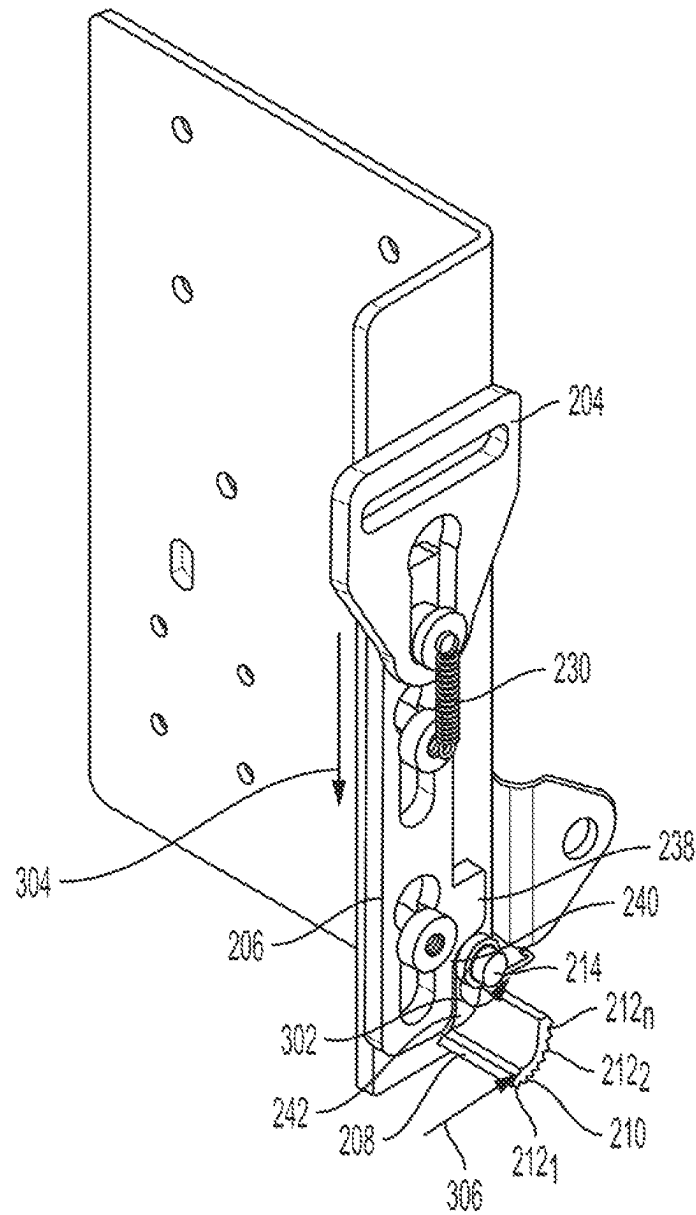
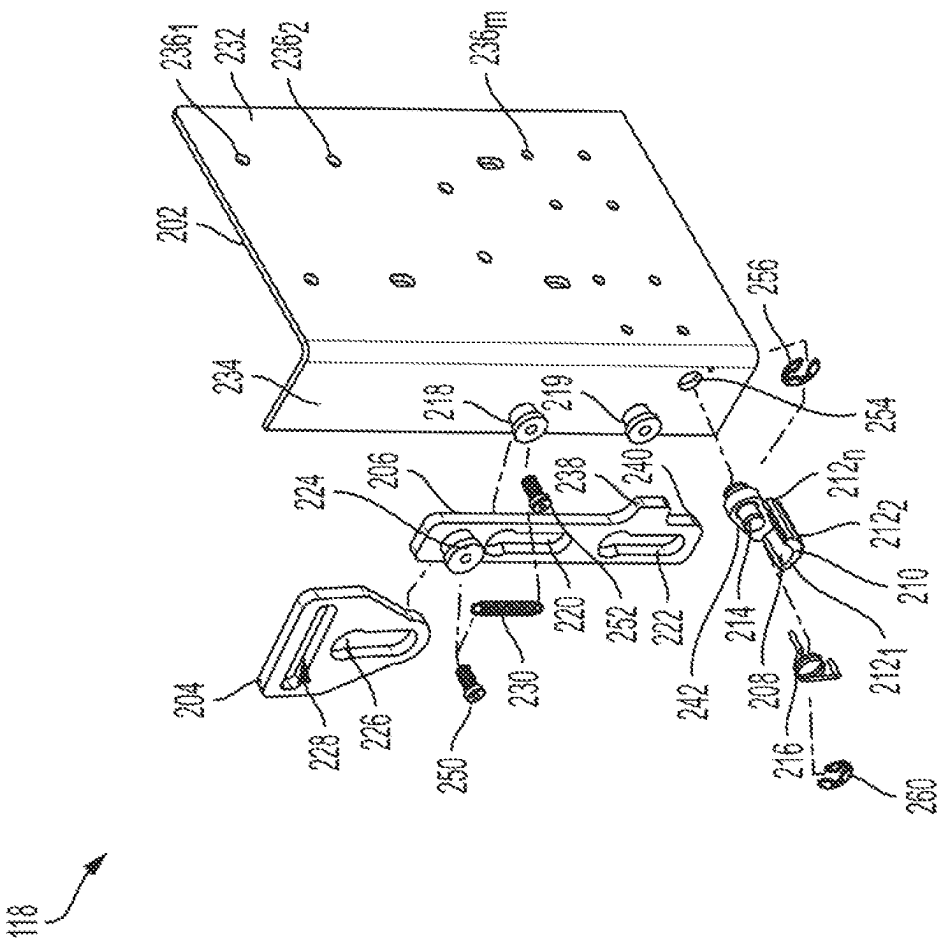


FIG. 3



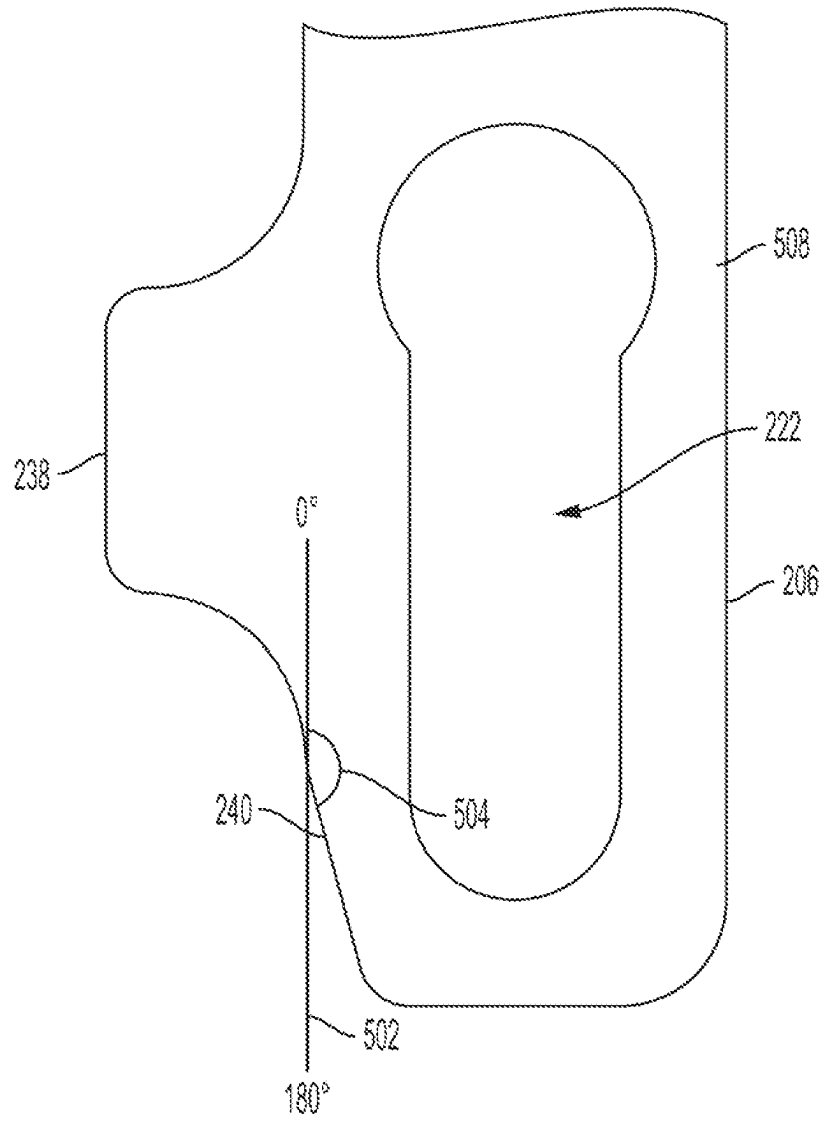


FIG. 5

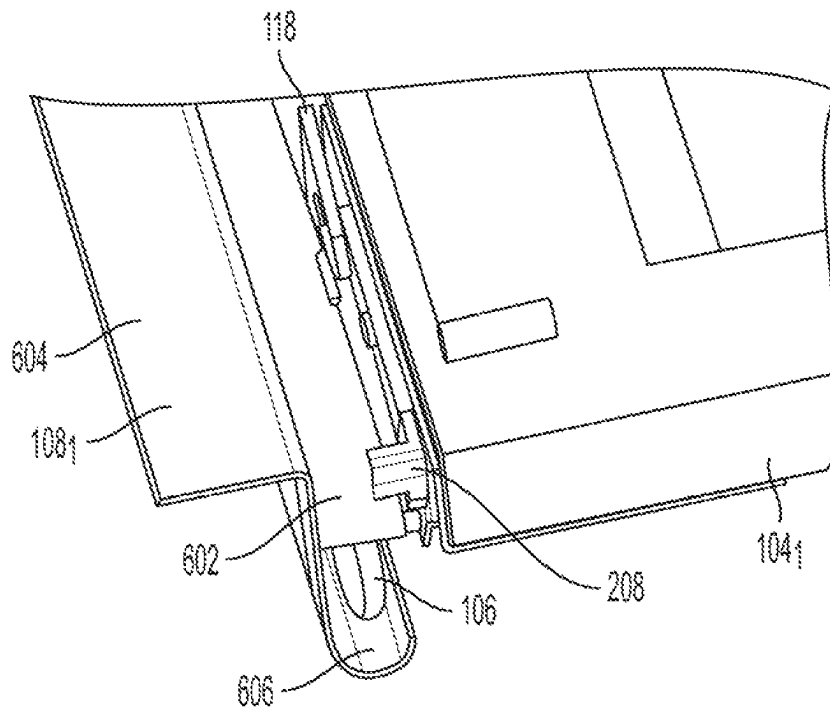


FIG. 6

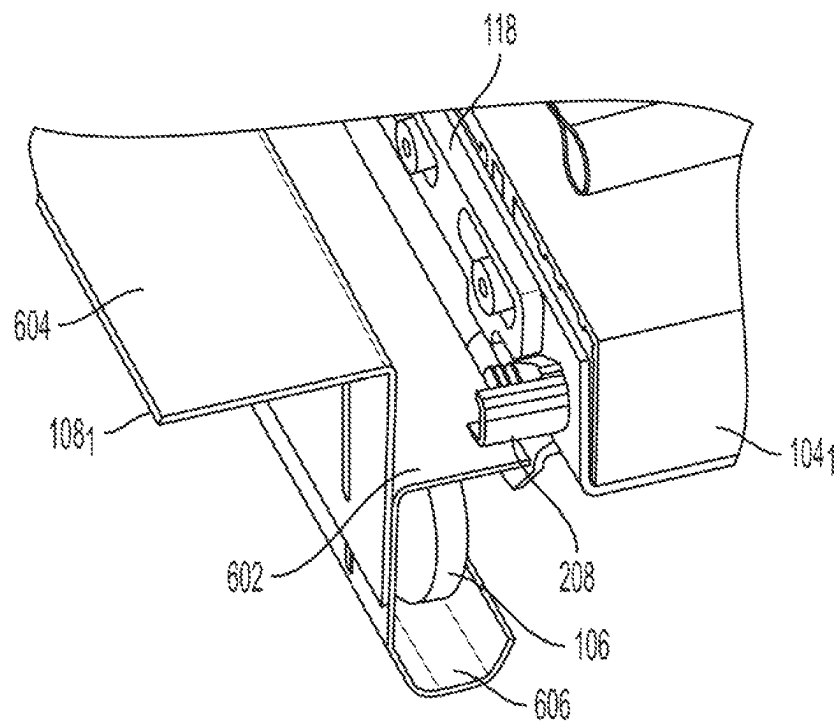


FIG. 7

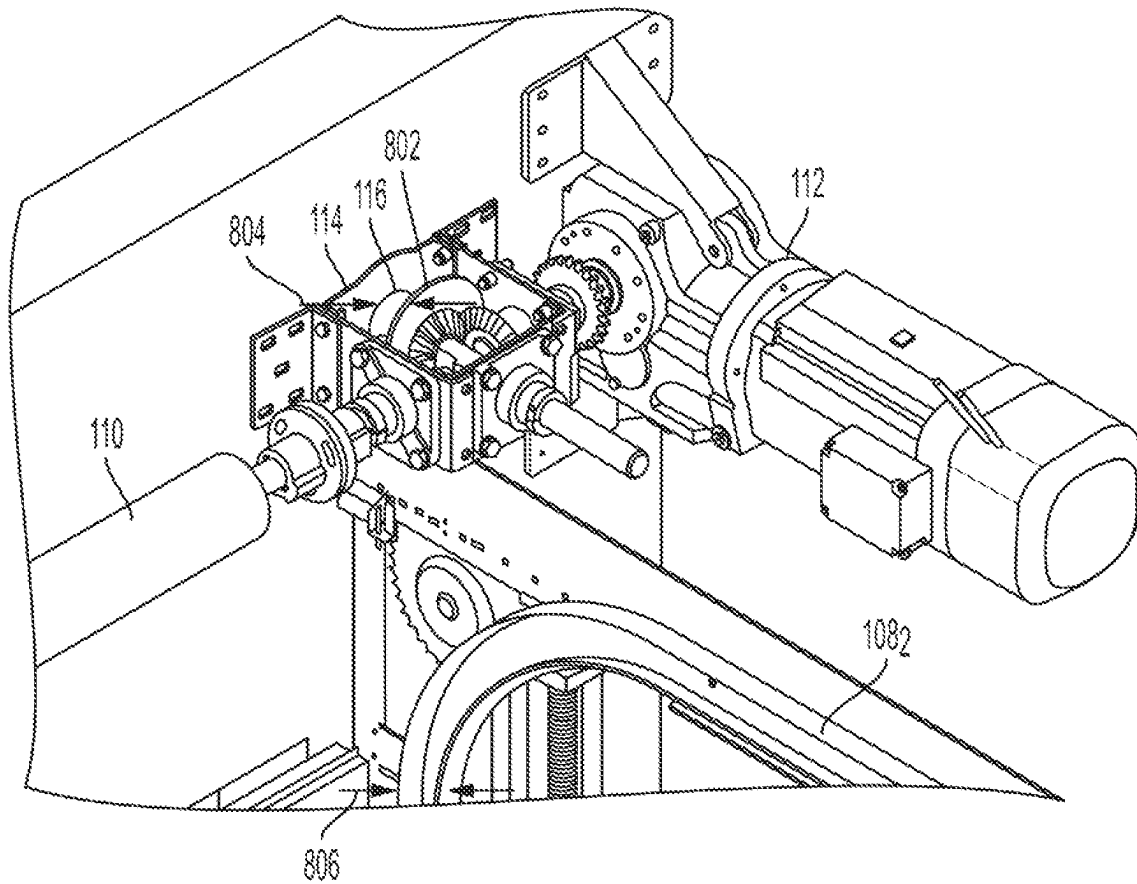


FIG. 8

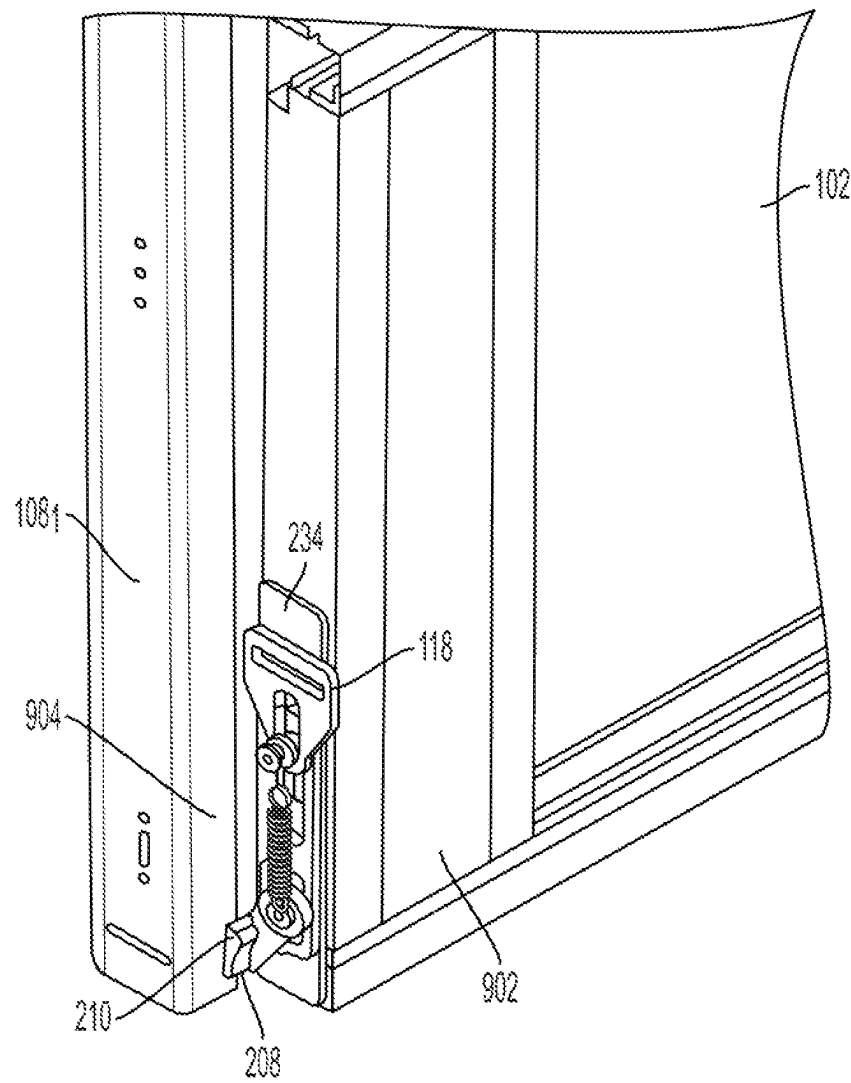
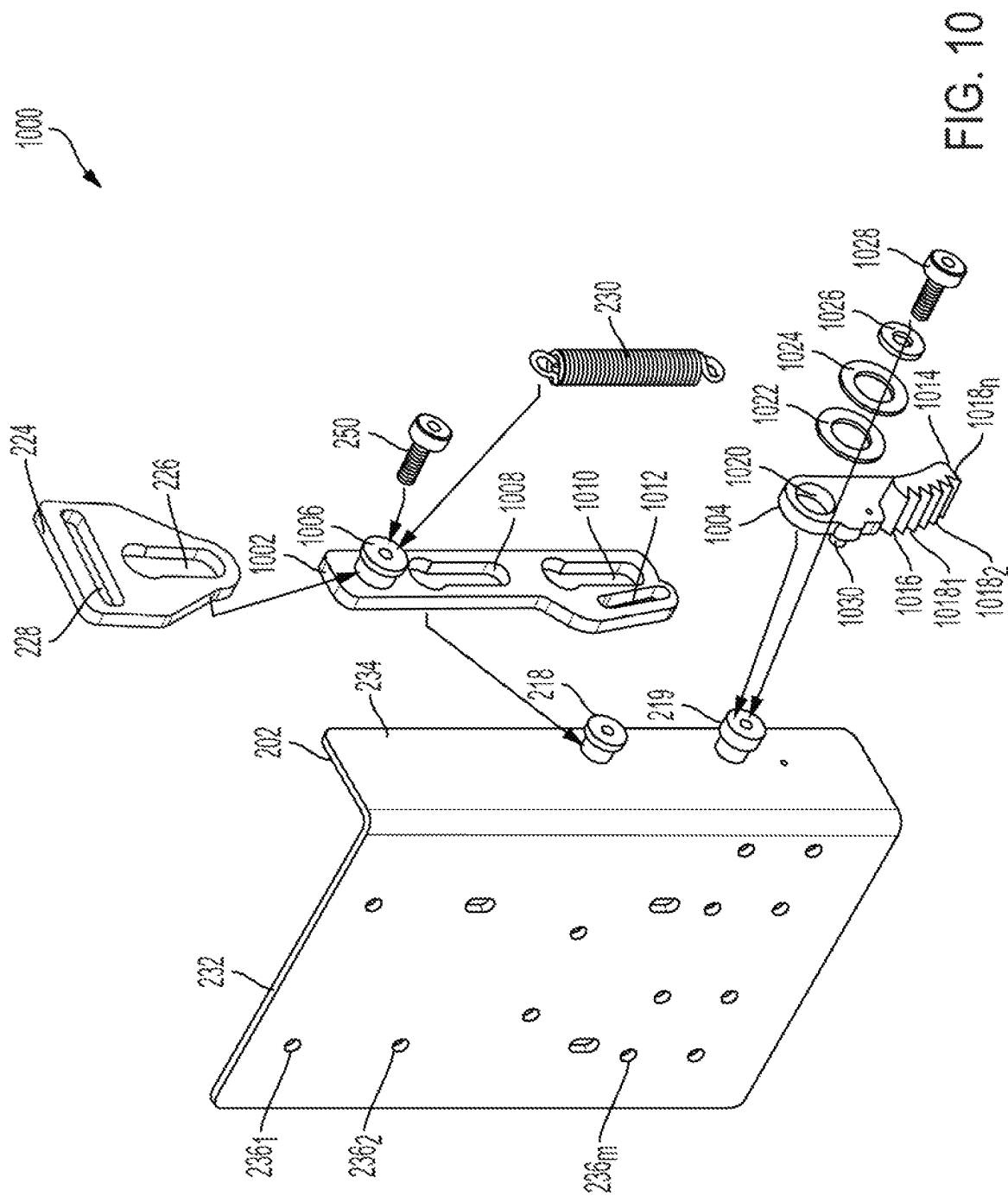


FIG. 9



1

DOOR STOP MECHANISM**BACKGROUND**

Overhead doors to cover external and internal openings may comprise panel sections that are connected by hinges. An overhead door can be raised into an open position and lowered to a closed position through rollers in a track system. The track system has a vertical section mounted to a wall, and a horizontal section mounted to a ceiling or overhead structure. In addition to the track system, a cable or strap system interacts with a counterbalance system comprising a torsion spring bar. The torsion spring bar comprises a torsion spring and cable drums located on each end of the torsion spring bar. The cable drum is connected to a first end of a cable or strap, and the second end of the cable or strap is attached to a bottom bracket, which is mounted to the bottom-most panel of the door.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of an overhead door with an example door stop mechanism of the present disclosure;

FIG. 2 is an isometric view of an example door stop mechanism in a disengaged position of the present disclosure;

FIG. 3 is an isometric view of the example door stop mechanism in an engaged position of the present disclosure;

FIG. 4 is an exploded view of the example door stop mechanism of the present disclosure;

FIG. 5 is a side view of a bracket of the example door stop mechanism of the present disclosure;

FIG. 6 is an isometric view of the example door stop mechanism of the present disclosure on a panel of the overhead door relative to a track and in a disengaged position;

FIG. 7 is an isometric view of the example door stop mechanism of the present disclosure on a panel of the overhead door relative to a track and in an engaged position;

FIG. 8 is an isometric view of an example gear box with a member that is attached to a strap connected to the example door stop mechanism of the present disclosure;

FIG. 9 is a view that illustrates how the door stop mechanism is coupled to a door; and

FIG. 10 is an exploded view of another embodiment of the door stop mechanism.

DETAILED DESCRIPTION

Examples described herein provide examples of a door stop mechanism for an overhead door. As discussed above, an overhead door can include a cable or strap system to help open and close the panels of the overhead door. One issue with cable or strap systems for overhead doors is that the tension of the cable or strap should be consistent so as not to cause misalignment or prohibit movement of panels of the overhead door when opened or closed.

In some instances, when the door is closing, there may be a point where the door is in a vertical section of the track and moved by gravity. Only the strap being controlled by the torsion spring bar controls the descending velocity of the overhead door. An uncontrolled descending velocity of the moving panels of the overhead door may be undesirable, and reliable means to arrest the movement of the overhead door is therefore desired.

2

The present disclosure provides a door stop mechanism to stop movement of the overhead door when tension in the strap is lost. For example, while tension in the strap is maintained, the door stop mechanism may be disengaged to allow the door to move freely within the track system. When tension in the strap is lost, the door stop mechanism may be moved into an engaged position to prevent the door from moving until the strap is replaced or tension in the strap is restored.

In an example, the door stop mechanism may include a braking system that engages the track. The braking system may use friction to “catch” the track and prevent movement of the overhead door.

The door stop mechanism of the present disclosure may provide a low-cost solution to arresting movement of overhead doors when tension in the strap is lost. The door stop mechanism of the present disclosure may also be retrofitted to existing overhead doors and provide a relatively low profile and clean design.

FIG. 1 illustrates an interior front view of an overhead door system 100 with door stop mechanism 118 of the present disclosure. FIG. 1 illustrates a view of the overhead door system 100 with a door 102 in the closed positioned.

In one embodiment, the overhead door system 100 includes the door 102. The door 102 may be comprised of a plurality of panels 104₁ to 104_n (hereinafter also referred to individually as a panel 104 or collectively as panels 104). The panels 104 may be vertically arranged. In some embodiments, the panels 104 may be movably coupled via panel fasteners or hinges (not shown) or may be disconnected from one another.

The panels 104 may be constructed of the same materials or different materials. In one embodiment, each panel 104 may be a metal frame structure without a middle panel (e.g., an open panel). The metal frame structure may comprise at least two vertical stiles and at least two horizontal stiles connected at right angles. The metal frame may further comprise a third vertical stile, a fourth vertical stile, a fifth vertical stile, and a sixth vertical stile forming multiple middle areas defined by the metal frame structure. The middle areas of this metal frame can be open, may comprise a solid metal panel, may comprise an open metal structure (e.g., screen, grid, grate, woven metal structure, or the like), may comprise a polyacrylate panel that is clear or translucent, may comprise polymethylmethacrylate (PMMA) that is clear or translucent, may comprise a polypropylene panel that is clear or translucent, may comprise a glass panel that is clear or translucent, or may comprise a laminate structure that is intended to have impact resistance.

In one embodiment, the overhead door system 100 may include a track system that includes a first track 108₁ and a second track 108₂ that are oriented parallel to each other and located adjacent to the ends of each panel. The door 102 may be guided into an open and closed position via the first track 108₁ and the second track 108₂. For example, wheels 106 or other mechanical means (not shown) may be coupled to the ends of each panel 104. The wheels 106 may be fitted inside of the first track 108₁ and the second track 108₂. As the door 102 is opened and closed, the wheels 106 may travel along the first track 108₁ and the second track 108₂.

In one embodiment, the overhead door system 100 may include a shaft 110 that is located over the door 102. The shaft 110 may be coupled to a gear box 114 on one or both ends of the shaft 110, and to a motor 112. The gearbox 114 may be coupled to an end of the shaft 110. The motor 112 may cause the shaft 110 to rotate and assist in the movement of the door 102 between an open position and a closed

position. A drum or spool may be coupled to the ends of the shaft **110** or coupled to the gear box **114**.

In one embodiment, a connection member **116** may be coupled to a portion of the spool or drum (shown in FIG. **8**) and the door stop mechanism **118**. The connection member **116** may be a strap or a cable. A strap may be fabricated from fabric or metal and may have a flat or ribbon shape. A cable may be fabricated from steel and have a cylindrical shape.

In one embodiment, FIG. **1** illustrates an example where the connection member **116** is a strap (hereinafter also referred to as the strap **116**). The strap **116** may be oriented such that a width of the strap **116** is parallel to a width of the track **108₁** and/or **108₂**. The drum may be selected to be concentrically aligned with the shaft **110**, allowing for the spooling of the strap **116**, or may be a gear box that allows for the strap to be oriented parallel to a width of the track **108₁** and/or **108₂**. Further details of how the strap **116** is coupled to the drum and the door stop mechanism **118** are discussed in further detail below.

Although a single door stop mechanism **118** and single strap **116** are illustrated in FIG. **1**, it should be noted that the overhead door system **100** may include two door stop mechanisms **118** and two straps **116**. For example, a door stop mechanism **118** and a strap **116** may be deployed on each end of the door **102**.

In one embodiment, the doorstop mechanism **118** may be positioned only on a bottom most panel **104₁**. As discussed in further detail below, the tension in the strap **116** may control whether the door stop mechanism **118** is engaged or disengaged. When tension in the strap **116** is lost, the door stop mechanism **118** may move to an engaged position that arrests movement of the door **102** or arrests movement of the panels **104** within the tracks **108₁** and **108₂**. When tension in the strap **116** is maintained, the door stop mechanism **118** may move to a disengaged position that allows door **102** to move freely within the tracks **108₁** and **108₂**.

In one embodiment, the strap **116** may be tightened to have an initial amount of tension to keep the door stop mechanism **118** in a disengaged position. Tension in the strap **116** may be lost when the amount of tension becomes less than the initial amount of tension that was set in the strap **116**. Tension may be lost in the strap **116** over time due to stretching of the strap **116**, movement or slight separation of the panels **104**, misalignment of the door **102** within the tracks **108₁** and **108₂**, and the like.

FIG. **2** illustrates an isometric view of an example door stop mechanism **118**. In one embodiment, the door stop mechanism **118** may include a strap connection member **204**, a body **206**, and a brake **208**. The strap connection member **204**, the body **206**, and the brake **208** may be coupled together mechanically via a combination of protruding members **218**, **219**, and **224** inserted into slots **226**, **220**, and **222** and a spring **230**. The strap connection member **204**, the body **206**, and the brake **208** may be coupled to a bracket **202**.

In one embodiment, bracket **202** may include a first side **232** and a second side **234**. The first side **232** and the second side **234** may be coupled together at a right angle, perpendicularly, or at approximately 90 degrees. The first side **232** and the second side **234** may be formed from a single piece of metal or sheet metal. The first side **232** may include one or more openings **236₁** to **236_m** to couple the bracket **202** to a bottom most panel **104₁** of the door **102**, illustrated in FIG. **1**. A mechanical fastener (e.g., a screw, a nut and bolt, a pin, and the like) may be inserted into each of the openings **236₁**

to **236_m** and into corresponding openings (not shown) in the bottom most panel **104₁** to secure the bracket **202** to the bottom most panel **104₁**.

In one embodiment, the body **206** may be coupled to the second side **234** of the bracket **202**. For example, the second side **234** may include the protruding members **218** and **219**. The protruding members **218** and **219** may each have a cylindrical shape with a cylindrical body that has a smaller diameter than the cylindrical head. The body **206** may include slots **220** and **222**. The slots **220** and **222** may include a first portion that matches a diameter of the cylindrical heads of the protruding members **218** and **219**. The slots **220** and **222** may include a second portion that is smaller in diameter or width than the first portion that matches the diameters of the cylindrical bodies of the protruding members **218** and **219**.

The protruding member **218** may be inserted into the slot **220**, and the protruding member **219** may be inserted into the slot **222**. The cylindrical bodies of the protruding members **218** and **219** may move within the second portions of the slots **220** and **222**. The larger diameters of the cylindrical heads of the protruding members **218** and **219** may keep the bodies **206** secured against the second side **234** of the bracket **202**.

The body **206** may include a protruding member **224**. The protruding member **224** may have similar dimensions and features to the dimensions and features of the protruding members **218** and **219** (e.g., the cylindrical body and a larger cylindrical head). The strap connection member **204** may include a slot **226**. The slot **226** may have a similar size and similar features to the sizes and features of the slots **220** and **222** (e.g., a first portion with a larger diameter and a second portion with a smaller diameter).

The strap connection member **204** may be coupled to the body **206** by inserting the protruding member **224** through the slot **226**, similar to how the protruding members **218** and **219** are inserted into the slots **220** and **222**. The strap connection member **204** may be secured to the body **206** via a spring **230**. The spring **230** may be coupled to the protruding member **224** and to the protruding member **218** via a screw or other type of mechanical fastener.

In one embodiment, the protruding members **218**, **219**, and **224** may be aligned vertically on the second side **234** of the bracket **202**. In other words, centers of the protruding members **218**, **219**, and **224** may lie on a common vertical line that would run parallel to an edge of the second side **234**. The amount of movement of the strap connection member **204** and the body **206** may be defined by lengths of the slots **220**, **222**, and **226**.

In one embodiment, the spring **230** may be loaded to pull the strap member **204** downwards. In other words, the spring **230** may be loaded to move the strap connection member **204** closer to the protruding member **218**.

The strap connection member **204** may also include a slot **228**. The strap **116** may be fed through the slot **228**. As noted above, when tension is applied to the strap **116**, the tension in the strap **116** may act against the spring **230** and move the strap connection member **204** in an upward direction. In other words, the tension in the strap **116** may pull the strap connection member **204** away from the protruding member **218**. As a result, the bottom of the slot **226** may pull up against the protruding member **224** on the body **206** and move the body **206** upwards. The upward movement of the body **206** may move the brake **208** into a disengaged position.

When the tension in the strap **116** is lost, the force of the spring **230** may overcome the remaining tension in the strap

5

116 and cause the strap connection member 204 to move towards the protruding member 218. This causes the slot 226 to move lower, thereby allowing the slots 220 and 222 to also move lower against the protruding members 218 and 219, respectively. This may cause the body 206 to move in a downward direction. The downward movement of the body 206 may move the brake 208 into an engaged position. FIGS. 3, 6, and 7 illustrate further details of how the brake 208 is moved between the disengaged and engaged position.

In one embodiment, the brake 208 may be rotatably coupled to the second side 234 of the bracket 202. The brake 208 may include a pivot member 214 that is inserted into an opening on the second side 234 (shown in FIG. 4). The brake 208 may be secured via a torsion spring 214 that is wrapped around the pivot member 214. The torsion spring 214 may be loaded to cause the brake 208 to move away from the external surface of the track 108₁ or 108₂ into a disengaged position.

In one embodiment, the brake 208 may include an arm 242 that is coupled to the pivot member 214. The arm 242 may interact with a cam surface 240 of a cam 238 of the body 206. The cam surface 240 may be inclined or formed at an angle to properly interact with the arm 242, as illustrated in FIG. 5, and discussed in further detail below. As the body 206 moves upward and downward, the cam surface 240 may move away from the arm 242 or move against the arm 242. When the cam surface 240 does not contact the arm 242, the torsion spring 214 may cause the pivot member 214 and the arm 242 to rotate clockwise and to move the brake 208 into a disengaged position. When the cam surface 240 acts upon the arm 242, the cam surface 240 may apply a force that overcomes the force applied by the torsion spring 214. This may cause the pivot member 214 to rotate in a counter clockwise direction as the cam surface 240 moves further downward against the arm 242. The counter clockwise rotation of the pivot member 214 may cause the brake 208 to move to an engaged position.

In one embodiment, the brake 208 may have a curved outer braking surface 210. The surface 210 may be curved along a single axis of the surface. For example, the curved outer braking surface 210 may be formed by curving a rectangular surface around a horizontal axis or around the pivot member 214. Said another way, the curved outer braking surface 210 may appear as a section cut away from a curved surface of a cylinder.

The curved outer braking surface 210 may face the tracks 108₁ and/or 108₂. The curved outer braking surface 210 may include a plurality of contact points 212₁ to 212_n (hereinafter also referred to individually as a contact point 212 or collectively as contact points 212). The contact points 212 may provide multiple surfaces that can “catch” the surface of the track 108₁ or 108₂. Thus, the multiple contact points 212 ensure that the brake 208 can engage the surface of the track 108₁ or 108₂. The curved outer braking surface 210 allows more contact points 212 to gradually engage the surface of the tracks 108₁ or 108₂ to gradually stop the downward movement of the door 102.

In one embodiment, the contact points 212 may be formed with a series of alternating peaks and valleys or depressions. For example, the contact points 212 may be formed as a series of teeth or in a saw-tooth pattern. However, any pattern may be deployed for the contact points 212 that can be applied to the curved outer braking surface 210. For example, the contacts points 212 may be formed as a file textured surface with a cross-hatch pattern, or any other type of irregular surface that may create friction to help stop downward movement of the door 102.

6

FIG. 3 illustrates an example of movement of the strap connection member 204, the body 206, and the brake 208 when the door stop mechanism 118 is moved to an engaged position. For example, when tension is lost on the strap 116, the force of the spring 230 may pull the body 206 downward in a direction shown by an arrow 304 towards the protruding member 218 that is in a fixed position on the second side 234 of the bracket 202.

As the body 206 moves downward, the cam surface 240 of the cam 238 may begin to act upon the arm 242 of the brake 208. The cam surface 240 may be cut an angle to form an inclined surface that can cause the arm 242 to rotate around the pivot member 214 as shown by arrow 302. The force of the cam surface 240 acting upon the arm 242 may be greater than the force applied by the torsion spring 216 to cause the arm 242 to rotate.

The rotation of the arm 242 may move the brake 208 towards a surface of the track 108₁ or 108₂ in a direction shown by an arrow 306. The contact points 212 of the curved outer braking surface 210 may gradually begin to interact with the surface of track 108₁ or 108₂. The contact points 212 may “grab” or “catch” the surface of the track 108₁ or 108₂ to arrest or stop the downward movement of the door 102.

When tension in the strap 116 is restored, the force applied by the tension in the strap 116 may act against the spring 230 to pull the strap connection member 204 upward in a direction opposite the arrow 304. The upward movement of the strap connection member 204 may pull the body 206 in the upward direction such that the cam surface 240 moves away from the arm 242 of the brake 208. As a result, the force applied by the torsion spring 216 may allow the arm 242 to rotate in a clockwise direction that is opposite the arrow 302. The rotation of the arm 242 in the clockwise direction may move the brake 208 away from the surface of the track 108₁ or 108₂ in a direction opposite the arrow 306 into a disengaged position.

FIG. 4 illustrates an example exploded view of the door stop mechanism 118. FIG. 4 illustrates the bracket 202 comprising the first side 232 and the second side 234. As shown in FIG. 4, the second side 234 of the bracket 202 may include protruding members 218 and 219. FIG. 4 also illustrates an opening 254 on the second side 234 of the bracket 202.

The pivot member 214 may be inserted through the opening 254 and held by a clip 256. The torsion spring 216 may be looped around the pivot member 214 and held in place by a clip 260.

The body 206 may be coupled to the second side 234 of the bracket by inserting the protruding member 218 through the slot 220 and the protruding member 219 through the slot 222. The strap connection member 204 may be coupled to the body 206 by inserting the protruding member 224 through the slot 226. The spring 230 may be secured between the protruding member 224 and the protruding member 218. For example, a first screw 250 may be fed through a first loop on a first end of the spring 230 and into a threaded opening of the protruding member 224. A second screw 252 may be fed through a second loop on a second end (opposite the first end) of the spring 230 and into a threaded opening of the protruding member 218. The threaded opening may be located in a center of the protruding member 224 and 218.

FIG. 5 illustrates a close up view of the cam 238 and the cam surface 240. As noted above, the cam surface 240 may be cut or machined to have an inclined surface. In one embodiment, the inclined surface may be formed to have an

7

angle **504**. The angle **504** may be measured relative to an axis **502** that runs parallel to an edge **508** of the body **206**.

In one embodiment, the angle **504** of the cam surface **240** is set such that the body **206** may move smoothly in a downward direction. In addition, the angle **504** of the cam surface **240** may be set such that the inclined surface of the cam surface **240** gradually acts against the arm **242** of the brake **208** to rotate the arm around the pivot member **214**.

In one embodiment, the angle **504** may be greater than 90 degrees. In one embodiment, the angle **504** may be from about 155 degrees to 175 degrees. In one embodiment, the angle **504** may be approximately 165 degrees.

FIGS. **6** and **7** illustrate more detailed views of the brake **208** in a disengaged and in an engaged position, respectively. FIG. **6** illustrates the brake **208** in a disengaged position. In one embodiment, the track **108₁** may include a panel **604** that can be coupled to the wall of an opening where the overhead door system **100** is installed. The track **108₁** and/or track **108₂** may include a surface **602** that interacts with the brake **208** and a wheel track **606**. The wheel **106** of the panel **104₁** may travel within the wheel track **606** as the door **102** moves between the open and closed positions.

In the disengaged position, the brake **208** moves away from the surface **602**. In other words, tension on the strap **116** causes the body **206** to move away from the brake **208** and allows the torsion spring **216** to rotate the brake **208** such that the brake **208** does not contact the surface **602**.

FIG. **7** illustrates the brake **208** in the engaged position. When tension on the strap **116** is lost, the spring **130** may cause the body **206** to move downward against the brake **208**. The brake **208** may rotate towards the surface **602** as the cam surface **240** acts against the arm of the brake **208**. The contact points **212** of the brake **208** may engage or contact the surface **602** to stop movement of the door **102**. Further details of how the door stop mechanism is coupled to the door **102** are illustrated in FIG. **9**, and discussed below.

FIG. **8** illustrates a more detailed view of the gear box **114** with a drum **802**. In one embodiment, to provide a clean look, the strap **116** may be positioned such that a width **804** of the strap **116** is parallel to a width **806** of the track **108₂**.

In one embodiment, the drum **802** may include a surface within two circular guides to keep the strap **116** on the surface within the drum **802**. The strap **116** may be installed to wrap around the drum **802** and through the slot **228** of the strap connection member **204**. Thus, the strap **116** may be installed without adding any twists to the strap **116** that could cause the strap **116** to be tangled or to get stuck in the slot **228** during movement of the door **102**.

FIG. **9** illustrates a view of how the door stop mechanism **118** is coupled to the door **102**. For example, the bracket **202** may be coupled to an inner side of the door **102**. FIG. **9** illustrates a view of an exterior side **902** of the door (e.g., the side of the door that faces the outside of the building).

The door stop mechanism **118** may be positioned such that the curved outer braking surface **210** contacts an exterior surface **904** of the first track **108₁** and/or the second track **108₂** when the brake **208** is in an engaged position.

FIG. **10** illustrates an exploded view of another embodiment of the doorstop mechanism **1000**. In one embodiment, the door stop mechanism **1000** may include the bracket **202** that is similar to the bracket **202** of the door stop mechanism **118** illustrated in FIG. **4**. In one embodiment, the door stop mechanism **1000** may also include the strap connection member **204** similar to the door stop mechanism **118**.

However, the door stop mechanism **1000** may include a body **1002** and a brake **1004** that are designed differently

8

than the body **206** and the brake **208** of the door stop mechanism **118**. In one embodiment, the body **1002** may include a protruding member **1006**. The slot **226** of the strap connection member **204** may be fitted over the protruding member **1006**. The spring **230**, secured by the screw **250**, may interact with the body **1002** and the strap connection member **204** by connection to the protruding member **1006** and the protruding member **219**.

In one embodiment, the body **1002** may also include slots **1008** and **1010**. The protruding member **218** may be inserted through the slot **1008** and the protruding member **219** may be inserted through the slot **1010**.

In one embodiment, the body **1002** may include a guide slot **1012**. The guide slot **1012** may interact with the brake **1004** to help guide movement of the brake **1004** between an engaged position and a disengaged position as the body **1002** moves up and down. For example, tension on the strap **116** may cause the body **1002** to move upwards and allow the guide slot **1012** to rotate the braking surface **1016** of the brake **1004** away from the exterior surface **904** of the track **108₁** and/or track **108₂**. When tension is lost on the strap **116**, the lost tension may cause the body **1002** to move downwards and allow the guide slot **1012** to rotate the braking surface **1016** of the brake **1004** towards the exterior surface **904** of the track **108₁** and/or track **108₂**.

The guide slot **1012** may have a width that is equal to or slightly larger than a diameter of a protruding member **1030** on the brake **1004**. The guide slot **1012** may have a length that allows the brake **1004** to move between the engaged and disengaged position. The guide slot **1012** may be set at an angle relative to the vertical up and down movement of the body **1002** to translate a linear motion of the guide slot **1012** into a rotational movement of the brake **1004**.

In one embodiment, the brake **1004** may include an opening **1020**. The brake **1004** may be positioned by inserting the protruding member **219** through the slot **1010** of the body **1002** and the opening **1020** of the brake **1004**. The protruding member **1030** on the brake **1004** may be inserted into the guide slot **1012**.

A screw **1028** may be inserted into the protruding member **219** to secure the brake **1004** against the body **1002** and the bracket **202**. In one embodiment, one or more spacers **1022** and **1024** and a washer **1024** may be inserted between the opening **1020** and the screw **1028**.

In one embodiment, the brake **1004** may include a braking surface **1016** that is similar to the braking surface **210**. For example, the braking surface **1016** may include a textured surface (e.g., teeth **1018₁**-**1018_n**, illustrated in FIG. **10**) or any other type of textured surface such as a filed textured surface with a cross-hatch pattern.

Thus, the present disclosure provides a mechanical door stop mechanism **118** or **1000** that can be deployed to stop movement of the door **102**. As noted above, although the door stop mechanism **118** and **1000** are shown on a single side of the bottom most panel **104₁**, it should be noted that the door stop mechanism **118** or **1000** may be deployed on both sides or either side of the bottom most panel **104₁**. Thus, the door stop mechanism **118** or **1000** may be deployed to interact with an outer or exterior surface **904** of the track **108₁** and/or track **108₂** where the overhead door system **100** is installed.

It will be appreciated that variants of the above-disclosed and other features and functions, or alternatives thereof, may be combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein

may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

The invention claimed is:

1. A door stop mechanism, comprising:
 - a bracket comprising a first side and a second side coupled together at a right angle, wherein the first side comprises one or more openings to be connected to a bottom most panel of an overhead door and the second side comprising a first protruding member and a second protruding member;
 - a body, comprising:
 - a first slot to receive the first protruding member of the bracket;
 - a second slot to receive the second protruding member of the bracket; and
 - a third protruding member, wherein the third protruding member, the first slot, and the second slot are aligned in a vertical direction;
 - a connection member, comprising a third slot to receive the third protruding member of the body;
 - a spring coupled to the body and to the connection member; and
 - a brake positioned adjacent to the body, wherein the brake moves between an engaged position via a spring force of the spring that causes the body to interact with the brake such that the brake contacts a track of the overhead door and a disengaged position via a force that acts against the spring force to pull the body away from the track to allow the overhead door to move.
2. The door stop mechanism of claim 1, wherein the connection member further comprises:
 - a slot to receive a strap.
3. The door stop mechanism of claim 2, wherein the spring is coupled to the third protruding member and the first protruding member.
4. The door stop mechanism of claim 1, the body further comprising:
 - a cam surface to engage a back side of the brake, wherein movement of the cam surface causes the brake to rotate between the engaged position and the disengaged position.
5. The door stop mechanism of claim 1, wherein the brake comprises:
 - a curved braking surface, wherein an outer side of the curved braking surface that faces the track of the overhead door comprises a plurality of contact points to contact the track.
6. The door stop mechanism of claim 5, wherein the plurality of contact points comprises teeth formed into the outer side of the curved braking surface.
7. The door stop mechanism of claim 1, wherein the brake is rotatably coupled via a torsion spring to the bracket.
8. The door stop mechanism of claim 1, wherein the body further comprises:
 - a guide slot to receive a brake protruding member on the brake.
9. The door stop mechanism of claim 8, wherein the brake comprises:
 - an opening to receive the second protruding member of the bracket; and
 - the brake protruding member to be inserted into the guide slot of the body.
10. A door stop mechanism, comprising:
 - a bracket comprising a first side and a second side coupled together at approximately 90 degrees, wherein the first

- side comprises one or more openings to be connected to a bottom most panel of an overhead door and the second side comprising a first protruding member and a second protruding member;
- a body, comprising:
 - a first slot to receive the first protruding member of the bracket;
 - a second slot to receive the second protruding member of the bracket; and
 - a third protruding member, wherein the third protruding member the first slot, and the second slot are aligned in a vertical direction;
- a connection member, comprising a third slot to receive the third protruding member of the body;
- a spring coupled to the body and to the connection member; and
- a brake positioned adjacent to the body;
- a drum; and
- a strap coupled to the drum and to the connection member, wherein tension applied to the strap causes the body to move away from the brake to move to a disengaged position such that the brake moves away from a track of the overhead door, and a loss in tension to the strap causes the body to interact with the brake to move the brake to an engaged position such that the brake moves towards the track and contacts the track.
11. The door stop mechanism of claim 10, wherein a width of the strap is parallel to a width of the body and the connection member.
12. The door stop mechanism of claim 10, wherein the body further comprises:
 - a guide slot.
13. The door stop mechanism of claim 12, wherein the connection member further comprises:
 - a slot to receive the strap.
14. The door stop mechanism of claim 12, wherein the brake comprises:
 - an opening to receive the second protruding member of the bracket; and
 - a brake protruding member to be inserted into the guide slot of the body.
15. An overhead door system, comprising:
 - a track system;
 - an overhead door movably coupled to the track system, wherein the overhead door comprises a plurality of panels;
 - a shaft located over the overhead door;
 - a drum coupled to a first end of the shaft;
 - a motor coupled to the first end of the shaft to rotate the shaft to open and close the overhead door and rotate the drum;
 - a door stop mechanism coupled to a bottom most panel of the plurality of panels;
 - a strap, wherein a first end of the strap is coupled to the drum and a second end of the strap is coupled to the door stop mechanism; and
 - a bracket coupled to the bottom most panel, wherein the bracket comprises a panel mounting surface and a door stop mechanism mounting surface coupled together perpendicularly, wherein the door stop mechanism mounting surface comprises a first protruding member and a second protruding member, wherein the door stop mechanism comprises:
 - a body, comprising a first slot to receive the first protruding member of the bracket, a second slot to receive the second protruding member of the bracket, a third protruding member, and a guide slot,

11

wherein the third protruding member, the first slot, and the second slot are aligned in a vertical direction;
 a strap connection member coupled to the second end of the strap, wherein the strap connection member comprises a third slot to receive the third protruding member of the body;
 a spring coupled to the body and the strap connection member; and
 a brake coupled to the second protruding member of the bracket and positioned adjacent to the body, wherein the brake comprises a brake protruding member that is inserted into the guide slot of the body, wherein tension applied to the strap causes the body to move away from the brake and causes the brake to rotate to a disengaged position such that the brake moves away from an exterior surface of the track system, and a loss in tension to the strap causes the body to move towards the brake and causes the brake to rotate to an engaged position such that the brake

12

moves towards the track system and contacts the exterior surface of the track system.

16. The overhead door system of claim **15**, wherein the brake comprises:

- a first end comprising an opening, wherein the second protruding member of the bracket is inserted through the opening; and
- a second end comprising a curved braking surface, wherein an outer side of the curved braking surface comprises a plurality of teeth to contact the track system.

17. The overhead door system of claim **16**, wherein a screw is inserted through the opening of the first end of the brake and into the second protruding member to secure the brake against the body and the bracket.

18. The overhead door system of claim **15**, wherein the guide slot is set at an angle relative to a vertical movement of the body to translate a linear motion of the guide slot into a rotational movement of the brake.

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